

Problem name: **Number Spiral (33)**

Summary of problem:

There will be given a row and a column of a matrix and the matrix is designed in such way which was displayed in question and we have to determine the value of the cell of that matrix.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int t;cin>>t;
    while(t--)
    {
        ll a,b;cin>>a>>b;
        if(a%2==0&&b%2==0)
        {
            a>b?cout<<a*a-b+1<<endl:cout<<(b-1)*(b-1)+a<<endl;
        }
        else if(a%2==0&&b%2!=0)
        {
            a>b?cout<<a*a-(b-1)<<endl:cout<<b*b-(a-1)<<endl;
        }
        else if(a%2!=0&&b%2!=0)
        {
            a>b?cout<<(a-1)*(a-1)+b<<endl:cout<<b*b-(a-1)<<endl;
        }
        else if(a%2!=0&&b%2==0)
        {
            a>b?cout<<(a-1)*(a-1)+b<<endl:cout<<(b-1)*(b-1)+a<<endl;
        }
    }
}
```

Output:

Input	Output
3	8
2 3	1
1 1	15
4 2	

Screenshot of Accepted Bar:

Task:	Number Spiral
Sender:	Arsenic
Submission time:	2023-09-06 19:18:31 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

It is an observation based problem where a cell position will be given and I have to output that value of that cell by doing some calculation. Time complexity and space complexity is $O(1)$.

Problem name: **Square in Circle (34)**

Summary of problem:

There will be given radius of a circle and have to determine the blank space of a square if we put in the perfect square.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
    for(int i=1;i<=t;i++)
    {
        double x;
        cin>>x;
        cout<<"Case "<<i<<": ";
        cout<<fixed<<setprecision(2)<<(((x*2)*(x*2))-2*acos(0.0)*(x*x));
        newl;
    }
}
```

Output:

Input	Output
3 20 30.091 87.0921	Case 1: 343.36 Case 2: 777.26 Case 3: 6511.05

Screenshot of Accepted Bar:

#	Submitted by	Problem	Language	Submitted at	Verdict	CPU	Memory
2918060	Asad Al Adil	Circle in Square	13 Days Ago	Judged	Accepted	11 ms	576 KB

Comments of mine:

It is a calculation based problem where I had to determine the area of the circle and the area of that square. After subtracting square area from circle area I got the result. As 2 digits had to be shown, I used setprecision to limit the decimal point. Time complexity and space complexity is $O(1)$.

Problem name: **Bit String (35)****Summary of problem:**

There will be given an integer number and we have to determine the total number of bit strings of that integer size.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    ll n,ans=1,mod=1e9+7;
    cin>>n;
    for(int i=1;i<=n;i++)
    {
        ans=(ans<<1)%mod;
    }
    cout<<ans;
}
```

Output:

Input	Output
3	8

Screenshot of Accepted Bar:

Task:	Bit Strings
Sender:	Arsenic
Submission time:	2023-09-06 20:34:45 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

As the input value is too big for that I had to use a loop to mod the actual value and I had used left shift to perform the operation. Time complexity is $O(n)$ and space complexity is $O(1)$.

Problem name: **Large Division (36)**

Summary of problem:

There will be given 2 numbers and we have to determine if the first number is divisible by second number or not. If divisible we have to print 'divisible' other wise 'not divisible'.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define mod0(x,y) x%y==0
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
    for(int z=1;z<=t;z++)
    {
        str r;
        cin>>r;
        ll x;cin>>x;
        ll ans=0;
        cout<<"Case "<<z<<": ";
        ll j=0;
        int n=r.size(),m=r[0]=='-'?1:0;;
        for(int i=m;i<n;i++)
        {
            ans=(r[i]-'0')+j;
            j=(ans%x)*10;
        }
        if(mod0(ans,x))
        {
            cout<<"divisible";
        }
        else
        {
            cout<<"not divisible";
        }
        newl;
    }
}
```

Output:

Input	Output
6 101 101 0 67 -101 101 7678123668327637674887634 101 11010000000000000000 256 -202202202202000202202202 -101	Case 1: divisible Case 2: divisible Case 3: divisible Case 4: not divisible Case 5: divisible Case 6: divisible

Screenshot of Accepted Bar:

#	Submitted by	Problem	Language	Submitted at	Verdict	CPU	Memory
2926520	Asad Al Adil	Large Division	5 Days Ago	Judged	Accepted	11 ms	576 KB

Comments of mine:

As first number could be 10^{200} , so I had taken the number in string and second one considered as an integer. Then I had done the simple calculation from first character to last character and taken the mod value of it for doing next calculation. Time complexity is $O(\text{string size})$ and space complexity is $O(1)$.

Problem name: **Power (37)**

Summary of problem:

There will be given 2 integer numbers m, n and we have to determine m^n .

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define ev(x) x%2==0
#define od(x) x%2==1
#define newl cout<<endl
const ll mod=1000000007;
int main()
{
    ll m,n;
    cin>>m>>n;
    ll x=1;
    while(n>0)
    {
        if(od(n))
        {
            x=(x*m)%mod;
        }
        n/=2;
        m=(m*m)%mod;
    }
    cout<<x;newl;
}
```

Output:

Input	Output
2 3 5 8	8 390625

Screenshot of Accepted Bar:

Lang	Status	Judge	Time	Memory	Code
C++	AC	20/20	00.00 s	3136 KB	1627 B

Comments of mine:

In my code I had run a loop until $n=0$ and multiplied the result value m to itself. But when n became odd I just multiplied it single time with the given number m . Time complexity is $O(\log n)$ and space complexity is $O(1)$.

Problem name: **GCD on Black Board (38)**

Summary of problem:

There will given n numbers and we have to find maximum gcd after moving a number between 1 to 10^9 .

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    ll b,ans=0;
    cin>>b;
    vector<ll>A(b+1),B(b+1),C(b+2);
    for(int i=1;i<=b;i++)
    {
        cin>>A[i];
        B[i]=__gcd(A[i],B[i-1]);
    }
    for(int i=b;i>=1;i--)
    {
        C[i]=__gcd(C[i+1],A[i]);
    }
    for(int i=1;i<=b;i++)
    {
        ans=max(ans,__gcd(C[i+1],B[i-1]));
    }
    cout<<ans;
}
```

Output:

Input	Output
3 7 6 8 3 12 15 18	2 6

Screenshot of Accepted Bar:

2023-09-22 20:57:51	C - GCD on Blackboard	Arsenic33Q	C++ 20 (gcc 12.2)	300	1803 Byte	AC	28 ms	5528 KB	Detail
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Comments of mine:

First of all I have calculated the gcd and stored in another array. Then reversely count the gcd and stored in another array serially. After that determined the maximum gcd from both of the arrays. Time complexity and space complexity is $O(n)$.

Problem name: **Beautiful Year (39)**

Summary of problem:

There will be given a year and we have to find another year from the next of the given year where every digits are unique.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    n++;
    while(1)
    {
        string s=to_string(n);
        map<char,int>mp;
        bool flag=true;
        for(int i=0;i<s.size();i++)
        {
            if(mp[s[i]]==0)
            {
                mp[s[i]]++;
            }
            else
            {
                flag=false;
                break;
            }
        }
        if(flag)
        {
            cout<<n;
            break;
        }
        else
        {
            n++;
        }
    }
}
```

Output:

Input	Output
1987 2013	2013 2014

Screenshot of Accepted Bar:

209549333	Practice: Arsenic33	271A - 11	GNU C++20 (64)	Accepted	30 ms	0 KB	2023-06-13 13:11:45	2023-06-13 13:11:45
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Comments of mine:

I had taken the number as integer and converted it into string and checked the count of every digit and searched for unique. Time complexity is $O((n_2 - n_1))$ and space complexity is $O(\text{string size})$.

Problem name: Least Common Multiple (40)

Summary of problem:

There will be given n integers number and we have to calculate the the least common multiple of those numbers.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    ll x=1;
    for(int i=1;i<=n;i++)
    {
        ll a;cin>>a;
        x=__lcm(x,a);
    }
    cout<<x;newl;
}
```

Output:

Input	Output
3	12
3 4 6	30
4	
1 2 3 5	

Screenshot of Accepted Bar:

Lang	Status	Judge	Time	Memory	Code
C++	AC	20/20	00.00 s	3112 KB	1555 B

Comments of mine:

In my code, I had run a loop for n integers and all the time calculate the lcm value with the previous counted lcm and then printed it. Time complexity is O(n) and space complexity is O(1).

Problem name: Raising Bacteria (41)

Summary of problem:

There will be given an integer number and we have to find the bacteria required to get that number as bacteria is growing 1 to 2 and 2 to 4.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    int x=0;
```

```

while(n)
{
    x+=(n%2);
    n/=2;
}
cout<<x;
}

```

Output:

Input	Output
5	2
8	1

Screenshot of Accepted Bar:

222545712	Practice: Arsenic33	579A - 18	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-09-09 08:55:22	2023-09-09 08:55:22
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Comments of mine:

In this case, I had just count the mod value after dividing by 2 until the number=0 and printed it. Time complexity is $O(\log n)$ and space complexity is $O(1)$.

Problem name: **Balanced Array (42)**

Summary of problem:

There will be given an integer number and we have to find a balanced array where the summation of the first half even number is equal to the summation of the second half odd number. If found then print 'YES' and the array otherwise only print 'NO'.

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define mod0(x,y) x%y==0
#define yess cout<<"YES\n"
#define noo cout<<"NO\n"
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
    while(t--)
    {
        ll n;
        cin>>n;
        if(mod0(n,4))
        {
            ll x=(n/2)*(n/2)+n/2;
            yess;
            for(ll i=2;i<=n;i+=2)
            {
                cout<<i<<" ";
            }
            for(ll i=1;i<=n-1;i+=2)
            {
                cout<<i<<" ";
            }
        }
    }
}

```



```

        x-=i;
    }
    cout<<x;endl;
}
else
{
    noo;
}
}
}

```

Output:

Input	Output
5	NO
2	YES
4	2 4 1 5
6	NO
8	YES
10	2 4 6 8 1 3 5 11
	NO

Screenshot of Accepted Bar:

224610514	Practice: Arsenic33	1343B - 11	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-09-22 15:59:42	2023-09-22 15:59:42
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Comments of mine:

First of all I had checked the number is divided by 4 or not. If not then printed 'NO'. Otherwise calculated the sum and firstly printed even numbers then printed the odd numbers using some looping condition. Time complexity is $O(n)$ and space complexity is $O(1)$.

Problem name: **Two Sets (43)**

Summary of problem:

There will be given an integer and we have to find if we can create 2 sets where the summation of the 2 sets are equal. If it is possible then print 'YES' and the 2 sets with size. Otherwise print 'NO'.

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define ev(x) x%2==0
#define od(x) x%2==1
#define yess cout<<"YES\n"
#define noo cout<<"NO\n"
#define newl cout<<endl
int main()
{
    ll n;
    cin>>n;
    ll x=(n*(n+1))/2;
    if(od(x))
    {
        noo;
    }
    else
    {
        yess;
        set<int>s1,s2;
        for(int i=1;i<=n;i++)

```

```

{
    s1.insert(i);
}
x/=2;
while(x)
{
    if(x>=n)
    {
        x-=n;
        s1.erase(n);
        s2.insert(n);
        n--;
    }
    else
    {
        s1.erase(x);
        s2.insert(x);
        break;
    }
}
if(s1.size()>s2.size())
{
    cout<<s1.size();endl;
    for(int i:s1)
    {
        cout<<i<<" ";
    }
    endl;
    cout<<s2.size();endl;
    for(int i:s2)
    {
        cout<<i<<" ";
    }
}
else
{
    cout<<s2.size();endl;
    for(int i:s2)
    {
        cout<<i<<" ";
    }
    endl;
    cout<<s1.size();endl;
    for(int i:s1)
    {
        cout<<i<<" ";
    }
}
}
}
}

```

Output:

Input	Output
7 6	YES 4 1 2 4 7 3 3 5 6 NO

Screenshot of Accepted Bar:

Task:	Two Sets
Sender:	Arsenic
Submission time:	2023-09-20 12:56:54 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

In this case, first of all I have checked the total summation is odd or even. If it even then inserted all the numbers in a set and erased one by one until the half of that summation and erased numbers were inserted into another set. For odd it is 'NO'. Time complexity is $(n \log n)$ and space complexity is $O(n)$.

Problem name: **Search II (44)**

Summary of problem:

There will be given n integers numbers and q queries. We have to find out the total number of how many numbers are there in the queries.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    map<int,int>mp;
    for(int i=1;i<=n;i++)
    {
        int a;
        cin>>a;
        mp[a]=1;
    }
    int q;cin>>q;
    int x=0;
    for(int i=1;i<=q;i++)
    {
        int a;cin>>a;
        if(mp[a]==1)
        {
            x++;
        }
    }
    cout<<x;newl;
}
```

Output:

Input	Output
5 1 2 3 4 5 3 3 4 1	3

Screenshot of Accepted Bar:

Lang	Status	Judge	Time	Memory	Code
C++	AC	10/10	00.06 s	10056 KB	1717 B

Comments of mine:

Here I have used map to count the total number and for queries I have just checked that if the map value is 1 or not. If 1 then count it. Time complexity is $O(n \log n)$ and space complexity is $O(n)$.

Problem name: **Distinct Numbers (45)**

Summary of problem:

There will be given n integers and we have to find out total unique number from it.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int n;cin>>n;
    set<int>s;
    for(int i=0;i<n;i++)
    {
        int a;cin>>a;
        s.insert(a);
    }
    cout<<s.size();
}
```

Output:

Input	Output
5 2 3 2 2 3	2

Screenshot of Accepted Bar:

Task:	Distinct Numbers
Sender:	Arsenic
Submission time:	2023-09-20 13:09:54 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

I have used set to find the number as set is stored unique numbers. After that I have just printed the size if it. Time complexity is $O(n \log n)$ and space complexity is $O(\text{unique numbers})$.

Problem name: **Apartments (46)**

Summary of problem:

There will be given n applicants and m free apartments. We have to determine the maximum number of applicant will get an apartment in their desired size.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define A_sort(x) sort(x.begin(),x.end())
#define newl cout<<endl
int main()
{
    int n,m,k;
    cin>>n>>m>>k;
    vector<int>A(n),B(m);
    for(int i=0;i<n;i++)
    {
        cin>>A[i];
    }
    for(int i=0;i<m;i++)
    {
        cin>>B[i];
    }
    A_sort(A);A_sort(B);
    int x=0,j=0;
    for(int i=0;i<n;)
    {
        if(abs(A[i]-B[j])<=k)
        {
            x++;j++;i++;
        }
        else if(A[i]<B[j])
        {
            i++;
        }
        else
        {
            j++;
        }
        if(j==m)
        {
            break;
        }
    }
    cout<<x;
}
```

Output:

Input	Output
4 3 5 60 45 80 60 30 60 75	2

Screenshot of Accepted Bar:

Task:	Apartments
Sender:	Arsenic
Submission time:	2023-09-20 13:43:23 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

First of all, I have sorted the numbers in increasing order. Then used 2 pointers to calculate the total value by checking the condition. Time complexity is $O(n \log n)$ and space complexity is $O(m+n)$.

Problem name: **Restaurant Customer (47)**

Summary of problem:

There will be given the arrival and leaving times of the n customers and we have to find out the maximum number of customers in a restaurant at any time.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    map<int,int>mp;
    for(int i=1;i<=n;i++)
    {
        int a,b;cin>>a>>b;
        mp[a]++;
        mp[b+1]--;
    }
    int a=-10000,b=0;
    for(auto it:mp)
    {
        b+=it.second;
        a=max(a,b);
    }
    cout<<a;
}
```

Output:

Input	Output
3 5 8 2 4 3 9	2

Screenshot of Accepted Bar:

Task:	Restaurant Customers
Sender:	Arsenic
Submission time:	2023-09-20 13:46:30 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

Here I have used map to determine the arriving and leaving times of the customers. For arriving +1 and leaving -1. Time complexity is $O(n \log n)$ and space complexity is $O(n)$.

Problem name: **Weird Algorithm (48)**

Summary of problem:

There will be given an integer n and if it is odd we have to $n=n*3+1$ and for even $n=n/2$ until $n==1$ and print the total numbers during this process.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    ll n;cin>>n;
    cout<<n<<" ";
    while(n!=1&& n>0)
    {
        if(n%2==0)
        {
            n=n/2;
            cout<<n<<" ";
        }
        else
        {
            n=n*3+1;
            cout<<n<<" ";
        }
    }
}
```

Output:

Input	Output
3	3 10 5 16 8 4 2 1

Screenshot of Accepted Bar:

Task:	Weird Algorithm
Sender:	Arsenic
Submission time:	2023-09-20 13:48:10 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:

In a while loop I have started the calculation as the problem statement and print the numbers until $n==1$. Space complexity is $O(1)$.